

Brahmins as scientists and science as Brahmins' calling: Caste in an Indian scientific research institute

Public Understanding of Science
2020, Vol. 29(3) 306–318
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DOI: 10.1177/0963662520903690
journals.sagepub.com/home/pus



Renny Thomas 
University of Delhi, India

Abstract

The domination of Brahmin and the upper caste scientists have given a Brahmanical identity to science in India. They have been perceived to be the natural inheritors of scientific practice, an assertion reaffirmed by scientists and researchers during my fieldwork in Bangalore, India. Furthermore, merit and passion for doing science was reinscribed and calibrated to denote the alleged castelessness and objectivity of science, obfuscating the deep hierarchies of caste in the practice of science in India. By using ethnographic methods, the article attempts to demonstrate how Indian scientists constructed their identities as casteless beings. The article calls for a public understanding of caste in Indian science, and suggests that public engagement has the potential to democratise the nature of science by addressing the question of exclusion and discrimination in Indian scientific institutions and universities.

Keywords

Brahmins, caste, culture, Dalits, laboratory ethnography, objectivity, reservations, science in India

1. Introduction

Even after the introduction of various inclusive policies such as reservations for marginalised groups in educational institutions, there is a stark absence of such groups in India's leading scientific research institutions. Available data show few scientists from the Dalit and other marginalised sections in most of the leading Indian scientific and technical educational institutions. This absence led me to undertake ethnographic fieldwork at the Indian Institute of Science (IISc), one of Asia's leading scientific research institutions, asking scientists from different disciplines and regions about questions of caste in science. This article, by studying caste and science, contributes to the existing scholarship on the various forms of bias and discrimination in science, such as race and gender.

Caste is a rigid form of social stratification in which one's position in society is solely based on the caste into which a person is born. The traditional hierarchical four-fold order consists of Brahmins (priestly caste), Kshatriyas (warrior caste), Vaishyas (trading caste) and Shudras (labouring caste).

Corresponding author:

Renny Thomas, Department of Sociology, Jesus and Mary College, University of Delhi, New Delhi 110021, Delhi, India.
Email: rthomas@jmc.du.ac.in

The group that was not part of the four-fold order was called ‘the untouchables’. They are called Dalits in modern India and are comprised of castes officially known as Scheduled Castes (SCs). However, this is just a simplified picture of the situation in India as there are more groups not fitting into the traditional caste structure outlined above. In modern India, various castes and groups that are socially and economically disadvantaged and excluded are called Other Backward Classes (OBCs). Compared to the SCs, comprising the lower castes among Hindus, Sikhs and Buddhists, OBCs also include some Muslim and Christian groups (see Mahajan, 2013).

With the reservation policy, the number of Dalits in educational institutions has increased, and they have begun to enter professions from which they were previously barred. But still their number is very small when it comes to various modern professions in India such as science, medicine, engineering and academia. Brahmins and other upper caste groups continue to dominate in all these professions and culturally shape the institutions of modern professions based on their caste identities and markers. In the case of Dalits, the association with degrading caste-based occupations generally prevents upwards mobility. For instance, Arunthathiyars, the Dalit caste group from the southern Indian state of Tamil Nadu, despite their rising education, are forced to work as manual scavengers, thereby ensuring a continued association with stigmatised work.¹

Though this article is an ethnographic study of one Indian scientific research Institution, it does talk to the data available from other scientific institutions and universities in India. Existing sociological studies demonstrate that there exist various forms of caste discrimination in many of the Indian universities, institutes and colleges (Chadha and Achuthan, 2017; Kondaiah et al., 2017; Pathania and Tierney, 2018; Tierney et al., 2019). Anthropological and sociological studies also confirm that caste-based discrimination and bias is not absent in leading Indian universities such as Indian Institutes of Technology (IITs) (Henry and Ferry, 2017; Rao, 2013; Subramanian, 2015, 2019). Historical studies show that the question of caste was never taken seriously in the discussions on Indian science. As Abha Sur (2011: 68) succinctly puts it,

The reluctance to acknowledge caste and class differences, perpetuated both by scientists involved and their biographers, serves to homogenize the scientific community in India, where achievements and failures are measured only in terms of individual potentials, effectively masking the institutionalized social relations of misrecognition.

2. Methodology and research questions

My attempt here is to demonstrate the ways in which caste works in a leading Indian scientific research institute, and the ways in which the upper caste scientists try to construct their identity as casteless beings, and how the Dalit and OBC scientists talked about bias and discrimination. This article presents ethnographic research that includes participant observation of a laboratory, and interviews with scientists and researchers from various departments such as Physics, Chemistry and Life Sciences. I was part of a laboratory at the Department of Chemistry for 1 year (February–December 2012). Though it was difficult to get access to be part of a laboratory as an ethnographer, the head of the laboratory, a distinguished scientist, helped me to get institutional affiliation as he was interested in social studies of science. Thus, I could live on the campus and interact with different sections. The laboratory I was associated with consisted of 35 members, who include the “Boss” (as the head of the lab was addressed by his students and researchers), visiting scientists, postdoctoral fellows, PhD students and research assistants. I interviewed 80 scientists from various departments of IISc in 2012 during my stay as a lab member. Also, in 2016, I interviewed selected scientists at IISc to ask specific questions on the dominant culture of the Institute, and also did participant observation of *Ayudha Puja*, a Hindu festival of worshipping tools, perceived as culture

by the scientists (Thomas and Geraci, 2018). This ethnography of caste in one scientific research institute allows us to see similar experiences in other institutions and locations. Names used are anonymized for ethical reasons.

3. Caste of science

Caste subordination is an everyday affair in leading institutions of science and engineering.² Indian newspapers narrate the experiences of Dalits, OBCs and minorities in the leading Indian scientific and technological universities such as IITs. There was a major controversy at IIT Madras over setting up separate entrances and wash basins for vegetarian and non-vegetarian students in the Halls. Students belonging to the Ambedkar–Periyar Study Circle³ termed it as untouchability (see Senthilair, 2018, for a detailed report of the incident). There are cases of ridiculing faculty members who come from reserved backgrounds and questioning their eligibility by the upper caste scientists. A case that caught the public's attention took place at the prestigious IIT Kanpur, where a Dalit faculty member was humiliated by his own colleagues who are upper castes, by making casteist remarks about him (Kaushal, 2018; *The Hindu*, 2019). There are reported cases of caste-based discrimination and bias in other IITs as well (see Mukherji, 2014).

IISc is the home of India's elite research scientists. Many of them are renowned scientists in their fields. Most of the scientists who work at IISc obtained their PhDs or postdoctoral training from leading American or British universities. Official data clearly show that the demography of IISc has changed little. It is still demographically and culturally dominated by Brahmin and upper caste scientists. The Annual Report of IISc 2017–2018 shows that IISc has a total number of 434 Academic Staff (Assistant Professors, Associate Professors and Professors). Out of 434 Academic staff, the Dalits and OBCs constitute only 18 (4.15%).⁴ Data from previous years also confirm the lack of representation of the SC/Scheduled Tribe (ST) and OBC scientists⁵ and show that they are not even 5% of the scientific community at IISc. Available data from other leading scientific and technical institutions such as IITs also show the absence of Dalits and OBC scientists.⁶ The ethnographic discussion follows very clearly confirms the official statistical data available for the IISc.

4. 'IISc as Iyer Iyengar Science Campus': Science as Brahmins' calling

In India, the generic scientist is still largely perceived to be Brahmin. The entry of non-Brahmin scientists has not changed this image. Many virtues of Brahmin culture have been normalised in the scientific institutional settings (Thomas, 2018; Thomas and Geraci, 2018). The domination of vegetarianism in India cannot be merely seen as food preference and choice as reports show that there are cases of discrimination on what one eats and cases reported on the existence of separate wash basins and entrances for vegetarians and non-vegetarians for example, as mentioned previously, in IIT Madras.⁷ The majority of the Brahmin scientists I spoke to at IISc were vegetarian. IISc has three major student dining halls (Mess), called A, B, and C: A mess is for pure-vegetarian food, B mess is for north Indian vegetarian and non-vegetarian food and C mess is for south Indian vegetarian and non-vegetarian food. I was curious to know about the composition of these halls. I joined in the C Mess as Boss had permitted me to be an apprentice in his lab and this allowed me to become a member at one of the Messes. I was intrigued by the differential mess arrangements. Why did they have a separate A Mess only for pure vegetarians when both B and C served vegetarian food of the same quality? Dileep, a non-Brahmin researcher who eats at C Mess, said once, 'A Mess is Brahmanical, it is meant for Brahmins'. Babu, another non-Brahmin scientist said, 'A mess is an Agraharam [a traditional Brahmin neighbourhood]'. Even though Brahmins and other upper

caste students and researchers ate at the B, and C mess, majority of the Dalits and non-Brahmins ate either at the B or at the C mess.

Many Brahmin scientists I spoke to informed me that they eat only vegetarian food. Some tried non-vegetarian food when they were in the Western universities. Food becomes a way of preserving their cultural and caste memory. Though these scientists continue to be vegetarians, they did not associate their food habits with their caste backgrounds. Instead, they justified vegetarian food habits as scientific. As one Brahmin scientist puts it,

Only now the West understood the value of vegetarianism. India already knew the value of vegetarianism and we knew that it is scientific.

This rationalisation of their food habits created the impression that other forms of food habits are unscientific. While this classification of scientific and unscientific dietary habits found no credence in science, it was used to justify this practice and thereby aligning it with modernity. Historian M. S. S. Pandian in his book *Brahmin and Non-Brahmin: Genealogies of the Tamil Political Present* captures this phenomenon among the Brahmins. He argued that the Tamil Brahmins presented the 'Brahmanical practices as always already modern' (Pandian, 2007: 37).

Due to their early entry into the Western education and their knowledge of English, it was the upper castes and Brahmins who benefitted, reshaped and negotiated these new forms of knowledge (Fuller and Narasimhan, 2015; Raj, 1991; Saberwal, 1979). They redefined the image of science as Brahmin and *Bhadralok* (educated upper caste Bengali) knowledge, which in turn was reaffirmed as the biography of science in India (for a discussion of Bengali *Bhadralok*, see Acharya, 1995). Much of this dominance from colonial times continue, due to which the survival of the non-Brahmin scientists was pegged on to their ability to conform to this identity. Brahmins set the norms, and others were expected to follow. This norm is visible from Sudarshan, a Tamil Brahmin scientist who works in the field of Aeronautics, when stating that the Brahmins are 'passionate' about learning, and others are interested only in 'money making'.

Speaking with Sudarshan at his office, I realised how the perception of science being a Brahmin profession was entrenched. I asked him why there were no non-Brahmin scientists or only few non-Brahmin scientists in the institutes of higher scientific learning such as IISc. He responded,

These guys are not interested to 'waste' time like us. They think it is better to go for a technical or business education, so that they would get immediate benefit. Science and research will not have an immediate effect. It takes time to get benefits. The most important thing one ought to have while pursuing research is 'patience', which we don't find among these groups. . .

Sudarshan did not think that he was offending anyone, but shared a common belief among many scientists that Brahmins are meant to do science and research, and that they are ready to sacrifice for knowledge. He narrated the case of a Muslim family he knew from his childhood:

Their son didn't study much, and he joined his father to do business. Now they have a successful business and doing very well. My father wasn't a very rich person, but he was very certain about our education. We had many problems, we didn't have proper food to eat, still we studied . . . I am not saying that only Brahmins would do this, but there is a high tendency among them. We value education, not on-the spot profit.

Sudarshan was not the only scientist who argued in this way. Many scientists with whom I interacted thought that learning and science was part of the Brahmin culture. Sudarshan's views reaffirm the alleged naturalness of Brahmins' intellectual superiority. It is they who valued education

and knowledge, and 'know the importance of education', and others are interested in 'on-the-spot profit'. John Harriss (2003: 332) in his analysis of Brahmin entrepreneurs in Chennai observes that they make it a point to argue that their parents were not rich, yet they ensured that their children got good education. Often, they find historical figures from their community to show this 'naturalness'. As Fuller and Narasimhan (2015) show, for many Tamil Brahmins, cultural figures were important to reproduce the 'natural' ability of Brahmins to excel in fields such as science and mathematics. Srinivasa Ramanujan, one of the most important cultural figures of the Tamil Brahmins, was born into a poor family and went on to become a world-renowned mathematician. When scientists like Sudarshan narrate their family background, they wish to convey that despite their poverty, they could study and become scientists as their family 'valued' education, and for them it is 'natural' that their family valued education.

This tendency to focus on their deprivation while ignoring the privileges of their Brahmin origin is observable throughout the personal writings and interviews with the Brahmin scientists. The redefinition and negotiation with Western education allowed them to not merely accept Western science, but accept it without compromising their cultural norms. This acceptance and negotiation made Brahmins the 'inheritors' of knowledge. It is this image which led Jagadesan, a Dalit scientist to tell me over coffee that *IISc means Iyer Iyengar Science Campus* (Iyers and Iyengars are Brahmins from the state of Tamil Nadu and settled in different parts of South India). Other non-Brahmin researchers I met, especially the Dalit researchers, also used 'Iyer Iyengar Science Campus' (IISc) to refer to the continuing hegemony of Brahmins in IISc. Ajantha Subramanian also notes in her study of IIT Madras that IIT is sardonically referred to by some in the vernacular press as 'Iyer Iyengar Technology' (Subramanian, 2015: 307).⁸

The notion that certain castes are 'naturally' fit to do some professions is prevalent among Indians. However, historical and anthropological studies show that this naturalness in the Indian context is historically wrong and that the upper castes and Brahmins dominate in various modern professions, including science, because of their access to modern education, and as beneficiaries of modern English education policy, they could enter various modern professions (Damodaran, 2008; Fuller and Narasimhan, 2015). Various sociological studies show that Brahmin practitioners of modern professions lived without compromising their cultural and traditional ways of life by compartmentalising their private and public life, or home and office (Khare, 1971; Singer, 1971), and historical studies show how the Brahmins adapted and transformed modern professions and science without losing their identity as Brahmins (Brown, 2012; Raina and Habib, 2004). They also invest time and resources in keeping alive their tradition and heritage (Harriss, 2003; Subramaniam, 2019; Thomas, 2019). As Brown (2012) argued, the *bhadralok* revealed a tendency to find the root of Western sciences in the Vedic scriptures and stressed the contribution of ancient India in the development of science (also see Nanda, 2010). In order to accept modern science, they had to give a new identity to it without compromising their cultural codes and norms. It is in this context that I talk about Prabhat, a scientist from IISc specialises in inorganic chemistry. He is a Tamil Brahmin, received his PhD from IIT Bombay before moving to the United States for his postdoctoral studies. In my conversation with Prabhat, he discussed the greatness and superiority of the Hindu tradition. He often recited Sanskrit verses and quoted the Hindu philosopher and saint Adi Shankaracharya to clarify his views on science. He narrated an incident that happened while he was doing his post-doctoral research to convey the superiority of Hindu tradition:

I am an early morning person. Usually I go to the Lab around 5.30 in the morning. I was walking to my lab when I met Jim. In the US it is very rare or almost never happens that somebody approaches you on the road. Generally, there are only few people one would see in the morning, as roads are not crowded as here in India. Suddenly one person approached me. He had a briefcase with him. He asked me, 'Hello sir, have

you seen God today'. I replied, 'Hello Sir, nice to meet you, yes I have seen God today'. He was taken aback and said, 'Look, no, no! I'm talking about God now', I said, 'Yes, I am also talking about God', he said, 'No sir, have you seen God today', 'Yes, I have seen', I said. He asked, 'Where, when, and how did you see God'. I said, 'I am just seeing God right in front of me'. He looked surprised, and responded, 'No, no, you do not understand. I am talking about God, not talking about me'. I said, 'Yes I understand. I am talking about God. And I am talking about you'. He asked, 'What do you mean?' Then he opened his briefcase and took a Bible and gave it to me. I said I have one. But he asked me to give him some time as he wanted to talk to me. I said I was busy and told him that I will give my address, and asked him to come to my place for a discussion. He was not very happy, and asked me if I had read Bible. When I said I had read Bible, he said, 'Then you are a Christian'. I said 'No'. Then I asked him to explain what God for him is. He could not answer. I informed him about Sankaracharya's definition of God and the philosophy of *neti-neti*. He asked me to explain. I told him, 'I conduct a study class. I teach the Vedas. It is called as Veda Prasang. People from different backgrounds come. You come over'. He came to the study group, and one day he said that he wanted to convert into Hinduism. I told him that it is the most ridiculous thing one can ever do. I conveyed to him that the idea of God is one, and Hinduism is not against any religion and God. Rather it respects all forms of God.

Prabhat was also proud of the openness of Hindu civilisation. Like many Brahmin scientists, he was also familiar with scriptures and he had also learnt classical Indian music. Prabhat also taught Vedas, ancient Hindu religious texts, during his stay in the United States. His pride in Indian civilisation comes from his Brahmanical upbringing. Therefore, when the evangelical preacher asked him about God, he discussed the philosophy of Sankaracharya, as he knew it as a Brahmin. By telling the preacher that converting into Hinduism is the most ridiculous thing one can ever do, he not only suggests that Hinduism does not support conversion, but also making a clear claim that Hinduism is the superior religion as it respects all religions. Many of the scientists I spoke to took pride in Hinduism but rarely mentioned about caste system in Hinduism. There is a noticeable trend among many Brahmin scientists to deny the existence of caste by saying that, 'I don't practice caste', 'what is caste?', 'I don't even know what it is', 'caste has no role in science', 'scientists don't believe in caste'. It is this idea of scientists as casteless which creates the false notion that caste has no place among scientists and scientific fields.

Caste, culture and science

The proficiency and knowledge in classical music and dance allowed the Brahmin and upper caste scientists to participate in various cultural programmes in IISc. The Brahmin scientists I interviewed enjoy classical music, especially Carnatic music (Hindu devotional music from the Southern part of India) in their free time. For instance, M. R. Iyer, an internationally renowned scientist and professor emeritus at IISc, had a good collection of CDs of Carnatic music in his office. He told me that he listens to Carnatic music when he is not working on scientific papers and research. Classical music and dance shape the Brahmin self. Anthropological accounts show that classical music is part of the everyday life of Brahmins (Visvanathan, 2014; Fuller and Narasimhan, 2015).

The classical forms of music and dance exclude the non-Brahmins. I realised the exclusiveness of classical music when I attended a Carnatic music concert at IISc organised by a student group. The singer was a renowned Carnatic musician. When she stopped singing, the audience continued to sing. They were well versed in the music she was singing and joined her without much effort. When it was time to ask her questions after the performance, I saw how well versed the audience were in their mastery of *ragas* (melodies in Carnatic music) from the innumerable questions that were posed to the singer. I attended several similar cultural programmes in IISc. What was

noticeable in all these programmes was the audience's familiarity with these art forms; their ability to understand and appreciate them and the exclusivity of the audience, which alienated the non-Brahmins. Scientists, who claimed to be atheists, also took part in these 'cultural' programmes (Thomas, 2017). They explained that they never thought these cultural programmes might alienate or exclude the non-Brahmins. The Brahmin scientists explained that classical music was part of their life and they kept their interest in music despite their busy life as scientists. They used their knowledge in classical music as a form of cultural capital to distinguish themselves from the non-Brahmins who did not possess the wherewithal to acquire classical music in their childhood.

It is to be noted that when Brahmin scientists tried to combine their taste for music and science, the non-Brahmin scientists were critical of this by showing either discomfort or lack of familiarity with the classical music as it is very clearly marked as an upper caste leisure form. Existing studies in the context of IITs show that students who come from SC/ST backgrounds are not attracted to leisure activities (Henry and Ferry, 2017: 12). Among various reasons, the lack of cultural capital of learning and knowing the dominant musical form is the major one. When the Brahmin scientists took pride in their musical abilities, the non-Brahmin scientists not only rejected it as Brahmanical, but also now arrive at counter cultures in places like IITs through Ambedkar–Periyar study groups. For example, when I asked about his opinion on various classical musical performances one sees on the campus, Babu, the non-Brahmin scientist said that he is not fond of 'classical' music. He liked popular film songs, folk songs and songs of resistance by Dalit reformers from Tamil Nadu. Babu said, 'classical meant Brahmanic and symbol of domination'.

Brahmin students from different backgrounds find IISc familiar, as they could continue many of their inherited traditions at IISc through various cultural associations. Ajantha Subramanian in her study of IIT Madras also observes that the Tamil Brahmin professors and students communicate in a language which can be understood only by them. The grammar of everyday conversation is familiar to both of them (Subramanian, 2015).

5. Caste, genealogy and scientists

The biographies and autobiographies of renowned Indian scientists surprisingly have no discussion about caste discrimination and inequalities. They write about their love for music, food and science. Autobiographies and biographies of scientists speak to us about their life, both professional and private, and as such they are as important as ethnographic data (see Fischer, 1995). In my reading of the autobiographies of three renowned Indian scientists (G Padmanaban, 2008; R Ramanna, 1991; CNR Rao, 2010),⁹ caste was a forgotten reality. C. N. R. Rao, one of India's most renowned scientists and a former director of the IISc, describes about his childhood and adulthood in a Brahmin household in the 1940s and 1950s without mentioning caste based discrimination and the practice of untouchability, which were prevalent then. He talks about his community without talking about its interaction with other castes.

G. Padmanaban, an internationally renowned biochemist and the former director of the IISc writing about his childhood and his father's profession, indirectly mentions that he belongs to a Tamil Brahmin community (Padmanaban, 2008). He talks about the reason why he wanted to pursue his research in biotechnology and agriculture. He stated that India being a poor country he wanted to contribute for the upliftment of poor people. Surprisingly, even when he talks about underdevelopment, poverty and malnutrition, he does not mention the Dalits who were the main victims of poverty and discrimination.

In his autobiography, Raja Ramanna, one of the founders of nuclear science in India, talks about everything except the caste inequality. He writes about his visit to various pilgrimage centres, and the greatness of Hindu civilisation.

These scientists also write about the educational qualifications of their ancestors. C. N. R. Rao's maternal grandfather was a teacher of Kannada and Sanskrit, and his father had a Master's degree. G. Padmanaban (2008) also talks about the educational qualifications of his parents and relatives. Ramanna writes about the genealogy of his caste group, the Hebbar Shri Vaishnava Brahmins of Karnataka. He writes with clarity and in detail about the history and legends of his caste group (Ramanna, 1991).

The ability to trace an illustrious genealogy and staking a claim to tradition and past is also a form of cultural capital. Clearly for the Dalits, the memory of stigma and caste oppression did not create a similar longing for the past. Many Brahmin scientists had role models at home, unlike the non-Brahmin and Dalit scientists, who were mainly first-generation learners. As Abha Sur (2011: 223) argues, 'The construction of illustrious genealogies is central to maintaining caste privilege. The caste system has achieved its endurance by laying claims to both heredity and socialization'.

6. Castelessness and caste capital

In this section, I will discuss how the Brahmin scientists' identity as casteless beings worked as a strategic tool to avoid questions of reservation and inclusion. Many scientists term their entry to science through 'passion', 'merit' and 'interest'. Being Brahmins who had educated family members, it was not just passion, but also their cultural capital that helped them in doing science. When I asked M. R. Iyer about reservation policy, he said,

I don't want to talk about it. Whatever you say will be very political. It has nothing to do with ideology. It is nothing to do with higher thinking. It is a purely political convenience. I have nothing to say. Let them do whatever they want. We must help people who are oppressed; there is no doubt about it. How to do is the question. Unfortunately, they must find a way. It may not be only reservation.

Similarly, Ramakrishna Bhat, a distinguished Neurologist, who hails from a Karnataka Brahmin background said,

I am reluctant to enforce affirmative action and reservation at a very high level, like PhD. I am not for it actually. I don't think it is required. I think affirmative action and reservation should be there actually up to 12th Standard, maybe you can include college, medicine, or engineering. After graduation, at the postgraduation level, I don't think it is required. For higher studies one requires interest, it is important to get good people who are interested in research. The researchers should be passionate, so we should be very careful about it, and one must be sure about it, otherwise it can destroy the system.

As Satish Deshpande (2013: 32) observed, 'Having encashed its traditional caste-capital networks and converted it into modern forms of capital like property, higher educational credentials and strongholds in lucrative professions, this section believes itself to be "caste-less" today'. This disavowal and denial of caste in their everyday life by claiming a liberal identity was common among the scientists of IISc. They would describe themselves as liberal, but when it comes to questions of reservation, they would be very sceptical. They believed, as Sudarshan said, in order to do science, one needs to have 'passion', and 'sacrificial mentality'. They believed that reservation may affect the quality of research, and reservation will produce scientists who are not passionate about science and research. Even the liberal progressive Brahmin scientists were sceptical about reservation. These scientists stated that caste should not be a criterion to get admission or recruitment.

Geetha Argade, a distinguished theoretical physicist, who belongs to the Saraswat Brahmin community of Maharashtra, told me that she does not even know the caste backgrounds of her

students, as for her 'all students are equal'. However, she, like other scientists, was sceptical about reservation, especially regarding faculty appointments:

Frankly speaking, I think at the highest level we should not have reservation. I don't think a professor at a university should be chosen because he or she belongs to a particular caste. There the criterion should be your merit and achievement.

When I asked Poornima, a renowned molecular biologist, who hails from a highly educated Brahmin household of Karnataka, on her views about caste and its presence in science, she was surprised to hear the word 'caste':

Caste!?? Oh, please! I have nothing to do with caste. It never registered in my mind. I will tell you; I was brought up in a very liberal way, and I have no concept of caste, religion, colour, creed etc. Every human being is equal in my eyes. I can't even relate to the concept of caste. If you are going to ask me questions of caste, I have no idea what to say. It doesn't even come to my mind.

When I asked her about the reservation policy, she said,

Oh, it is absolutely essential. But I do think after 60 years we should have overcome it, but I think it is so much entrenched by society and the politicians' inability to breakdown the barrier. The people are still for things like this, not the educated people, but the common man who votes, and that is again because of lack of political will.

Chandrasekharan, a distinguished biologist at IISc, who comes from a Nambudiri Brahmin (from Kerala) background was sceptical about reservation policy:

Certain people have vested interests in keeping caste alive. Because of identity business, you get something out of it. Reservation up to some level is affirmative action. By doing this, some sections have benefited. It has been mixed up with this identity politics and it has become difficult to talk about. It is important to make sure that it doesn't go to the wrong person. Some people continuously enjoy the benefit of reservation, and therefore the idea of creamy layer is very important. Reservation as one element in the whole affirmative action is necessary, but it should not be indefinite. But it has become a political thing. It is a very strange country where you want to be called 'Backward'. But yes, I would not decry reservation altogether. If it is properly used, absolutely carefully used and ensured that the well to do people in the caste would not monopolise it, then it is an important thing. But now it is part of politics and there is no way to deal with it.

These Brahmin scientists stated that to be a scientist, one must be 'good', 'efficient' and 'passionate'. They used these words uncritically to describe scientific practice. Merit and competence become objective in their everyday language, due to which questions of privilege and hierarchy are completely erased from their understanding of science. Reading these narratives provides us with an idea of how questions of caste were consciously silenced and elided in elite scientific institutions like IISc. Science and scientific practices were pegged on to the ability, merit and intelligence of the scientists and reservation was a hindrance for doing good science. For them, equality was about a level playing field and opportunities, which anyone irrespective of their caste backgrounds and the resultant social stigma could enter and make their name in science. However, the apparent castelessness and claims of objectivity among scientists coexisted with a significant interest in genealogies and family origins and in their preferences for distinct food habits and taste in music. When it came to questions of reservation, they were sceptical of it, claiming that it perpetuated

identity politics, vote bank politics, and so on. Here, it is important to remember that the number of academic scientists who belong to SC/ST and OBCs are less than 5% of those in IISc according to the official data.

At IISc, the scientists celebrate all the major Hindu festivals (Geraci, 2018, 2019; Thomas and Geraci, 2018). The dominant group has already shaped the cultural and intellectual atmosphere of the place. When I asked him about the cultural dominance of Brahmins, Babu, said, 'The atmosphere the Brahmins created continues, and will continue to exist'. As Bourdieu argues (1975: 29), 'The dominant are committed to conservation strategies aimed at ensuring the perpetuation of the established scientific order to which their interests are linked'. The culture of the Brahmins continues to be the culture of elite Indian Institutions such as IISc and IITs.

7. Conclusion: For a public understanding of caste in science

Though this article is an ethnographic study of an elite Indian scientific research institute, it is possible, with the help of available studies and data on other scientific and technological institutes, discussed above, to understand the prevalence of caste-based exclusions, domination and resistance in the context of India. The ethnographic discussion of this particular scientific institute along with other studies on the social life of caste among scientists invites us to think about the need for a public engagement with the caste question in science. Addressing caste questions in science will generate new debates about the nature of science among the public. Discrimination that happens in scientific institutions should be discussed with the public, who should be able to ask questions of caste, gender and racial discriminations in scientific institutions. In the context of India, public engagement could help to address the question of exclusion and discrimination in scientific institutions and laboratories. This article is an attempt in that sense to talk about caste in science having the public in mind.

Indian Sociologist Sharmila Rege (2013: 2) has noted, 'Since the upper castes have dominated urban middle-class arenas such as universities and research institutes, caste identity has hardly ever been an issue for public discussion' (also see Guru, 2001). However, with the Dalits and the OBCs entering higher education, there is an increased interest in understanding the everyday life of many of these elite scientific institutions. Media and newspapers carry report on these institutions, including matters such as caste questions in science. However, there is not much discussion on the nature of science among the public. Though there exist international studies on how the public perceive scientists and science (Prpić, 2011), and its relevance for an engaged science–society relationship, one does not find studies of that kind in the South Asian context, especially in the Indian context.

Only recently have the media started reporting on the underrepresentation of Dalits and OBCs in elite scientific research institutes. The public understanding of science and scientists will also change with the coming of new data on elite institutes like IITs and IISc. The public should be informed regularly regarding the number of Dalit, Tribal and the OBC scientists that they appoint, and various strategies that they employ to make the scientific and technical education more inclusive. If this debate around the underrepresentation of scientists does not happen, the elite institutes such as IISc and IITs will remain ivory towers and will not be accountable to the public.

The present ethnographic study of science and caste aims to contribute to the discussion of exclusion in sciences and, in that process, also aims to raise awareness of the culture of doing science in India. Beyond this ethnographic study of the caste question in Indian science, and building on data related to caste composition of Indian scientists, it will also be important to study how the public perceive caste in science. Therefore, this article invites future research on how the public perceive the question of caste and science, and how they perceive science and scientists in India.

Although the article is an ethnographic account of one scientific institution, as noted, it is very clear from the official statistical data and other studies available on elite scientific institutions that the caste-based discrimination, and domination of the upper castes, is a reality in India and needs wide discussion.

Acknowledgements

I am grateful to Susan Visvanathan (Jawaharlal Nehru University (JNU)) and Martin W. Bauer (London School of Economics and Political Science (LSE)) for reading previous drafts of this article and giving useful suggestions. I am deeply grateful to all the scientists and friends in Bangalore for their friendship and support, especially Boss. I thank the two anonymous reviewers and the editor for their very insightful comments that helped in revising the article. Special thanks to Susan Howard for patiently responding to all my queries.

Funding

The author(s) received no financial support for the research, authorship and/or publication of this article.

ORCID iD

Renny Thomas  <https://orcid.org/0000-0003-0845-3349>

Notes

1. For a discussion on the introduction of the Special Reservation Act and the consequent opposition from other Dalit castes, see Pandian (2013).
2. Leading Indian newspapers report on cases of discrimination faced by Scheduled Caste (SC) and Scheduled Tribe (ST) students and faculty members at Indian Institutes of Technology (IITs), see, for instance, Mukherji (2014), *The Hindu* (2019).
3. Student group based in IIT Madras to discuss the progressive ideas of Ambedkar and Periyar, two of the foremost anti-caste social reformers and thinkers of modern India.
4. Out of total 18, the number of SC/ST scientists are 10, and Other Backward Classes (OBCs) are 8 (Annual Report, IISc (Indian Institute of Science), 2017-2018: 11).
5. As per 2013–2014 IISc Annual Report, out of the total Academic staff of 410 (SC/ST=13, OBC=3, and General=394). The percentage of SC/ST and OBC scientists: 3.9% (Annual Report, IISc, 2013-2014: ix). The 2015–2016 report shows a total of 431 academic staff, and out of 431, the General Category constitute 413 numbers, SC/ST 9 and OBC 9: 4.18% (Annual Report, IISc, 2015-2016: 7). Here, I have only included the Academic Staff (Assistant Professors, Associate Professors and Professors). It is hard to get data on SC, ST and OBC scientists other than the data available from the Annual Reports, and there is no clarity on whether the 'General Category' also has scientists from these groups.
6. For instance, recent data reported by the Indian newspaper *The New Indian Express* show that the elite IIT-Madras has only 12.4% of faculty belong to reserved categories (see Chaitanya, 2019). Data available for other IITs are not different, for instance, IIT Kharagpur has three Dalit Professors and IIT Roorkee has only one Dalit Professor (see Mukul, 2012). The prestigious IIT Kanpur as per 2016–2017 has only three Dalit faculty members out of total 394 faculty members (see Kaushal, 2018). The Annual report of IIT Kanpur (2016–2017) shows that the institute has appointed 13 Academic Staff during 2016–2017, and all the 13 of them come from General category (see Annual Report, IIT Kanpur, 2016-2017).
7. Various leading Indian newspapers such as *The New Indian Express* (14 December 2018), and *The Hindu* (15 December, 2018) have reported this event. See Senthilir (2018) for a detailed report.
8. For a more historically grounded discussion on the categories of Brahmin and non-Brahmin, see Pandian (2007). He illustrates in detail the genealogies of the opposition between the categories of Brahmin and non-Brahmin in the context of Tamil Nadu (Pandian, 2007).
9. Both Rao and Padmanaban were directors of IISc. Raja Ramanna was the director of National Institute of Advanced Studies (NIAS), which is also part of the IISc campus, and has close academic and cultural relations with IISc.

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Author biography

Renny Thomas is an Assistant Professor in the Department of Sociology, Jesus and Mary College, University of Delhi, New Delhi, India. His research interests include science and religion, technology and rituals, rationality, scientific method, and laboratory life.